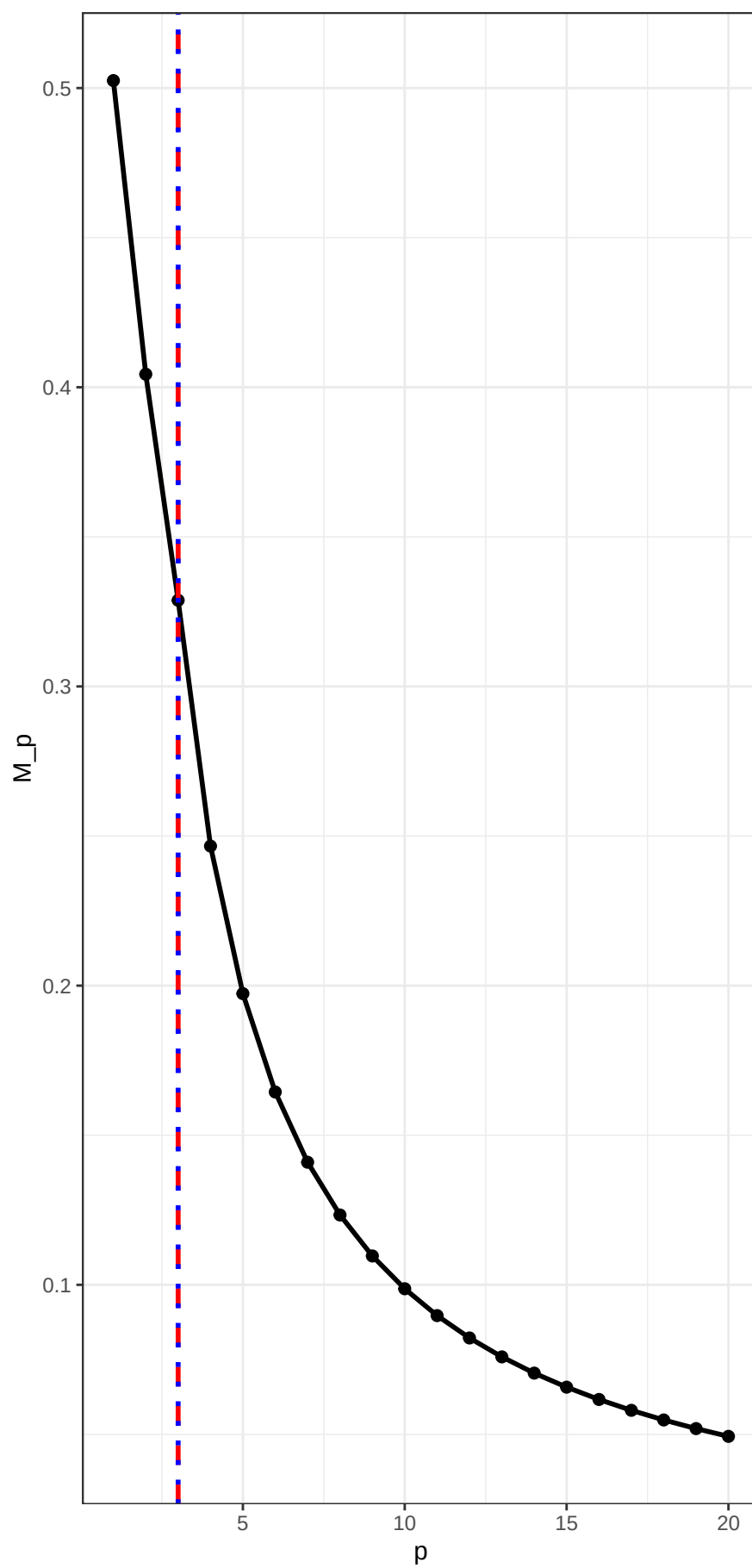


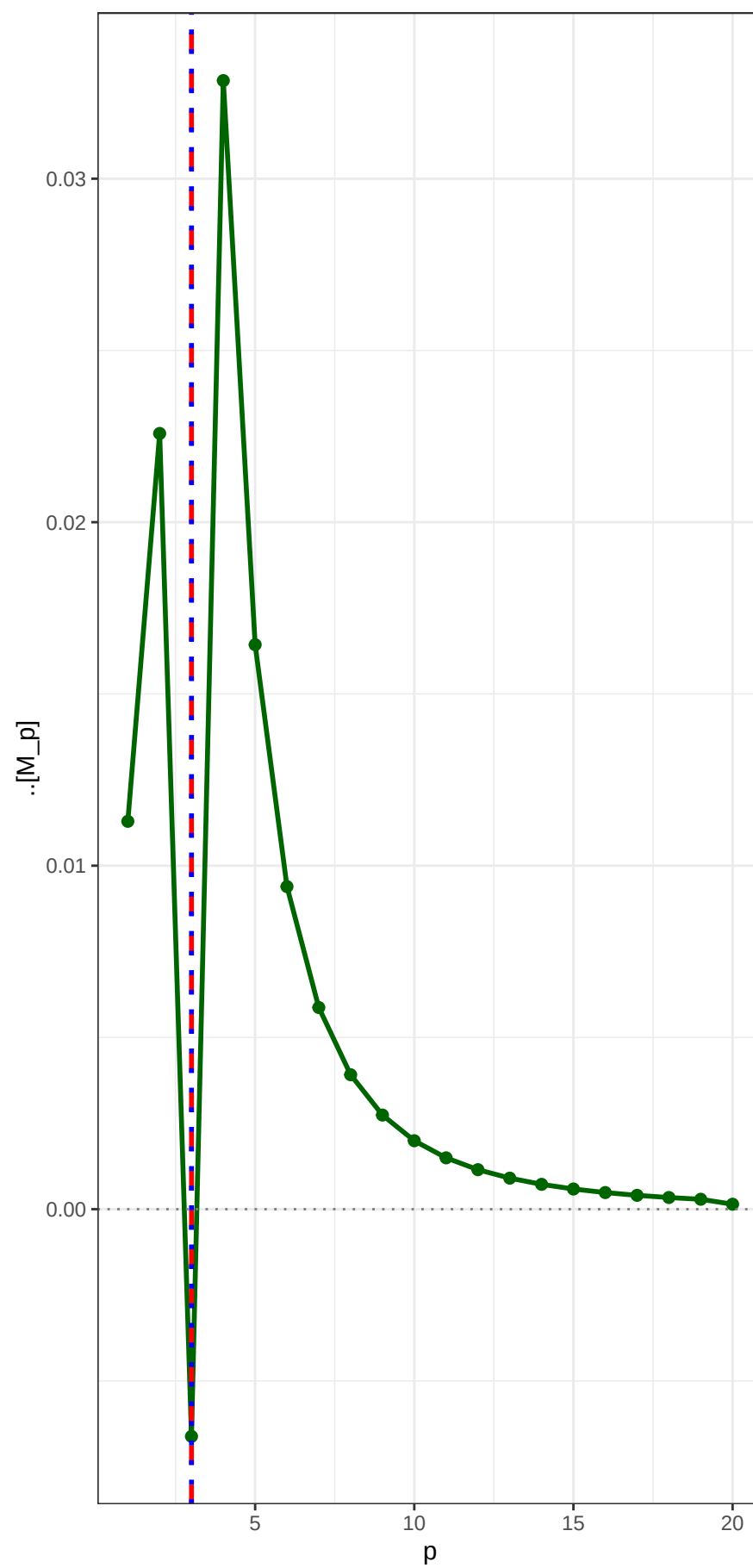
A1_Baseline_Uncorrelated: $M_p = R^2/p$

$p_{\text{true}}=3$ (red), $p_{\text{classical}}=3$ (blue)



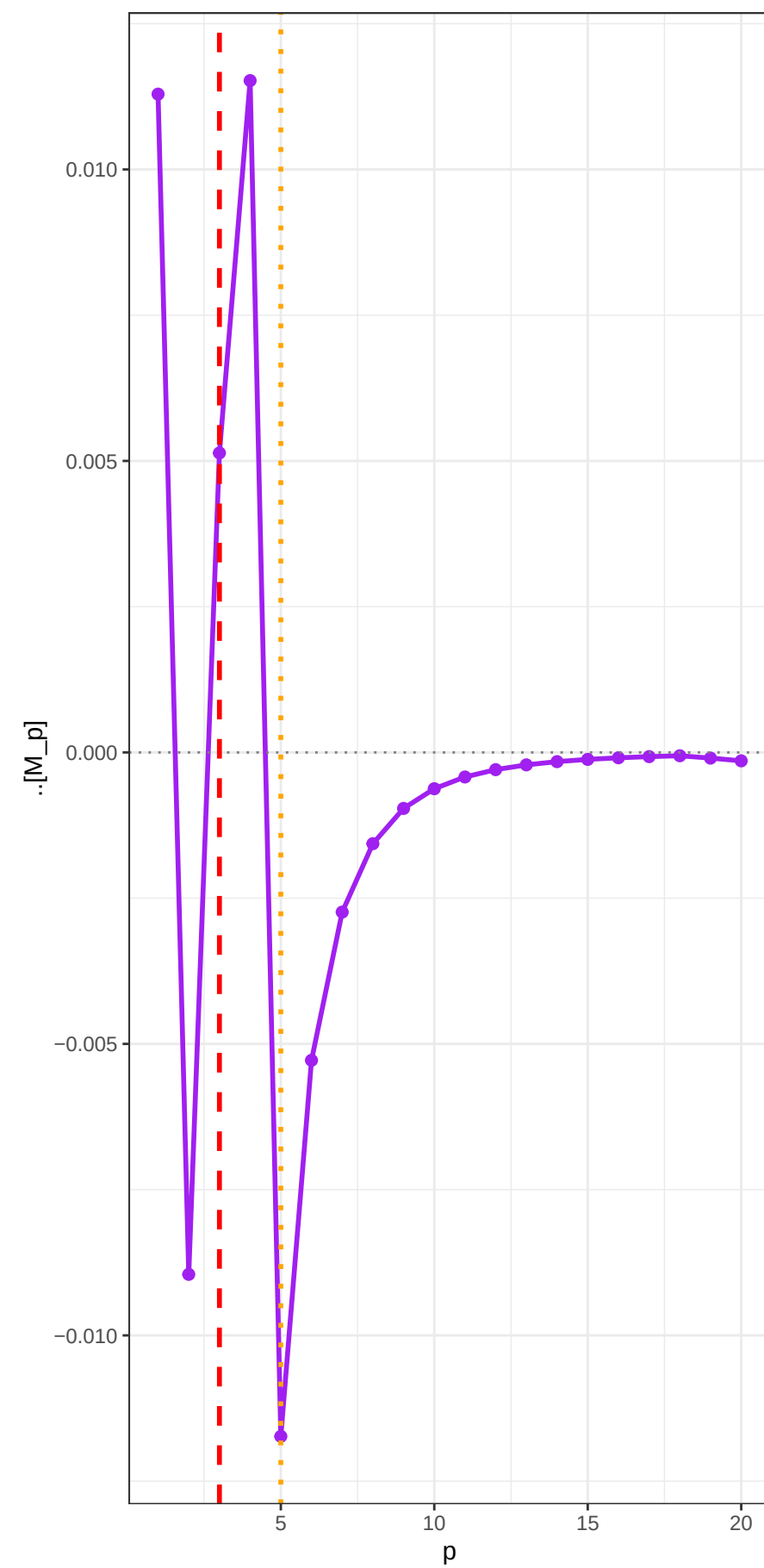
Second Derivative (Curvature)

Sign change: $\text{delta2}[2]=0.02258337 \rightarrow \text{delta2}[3]=-0.006614$



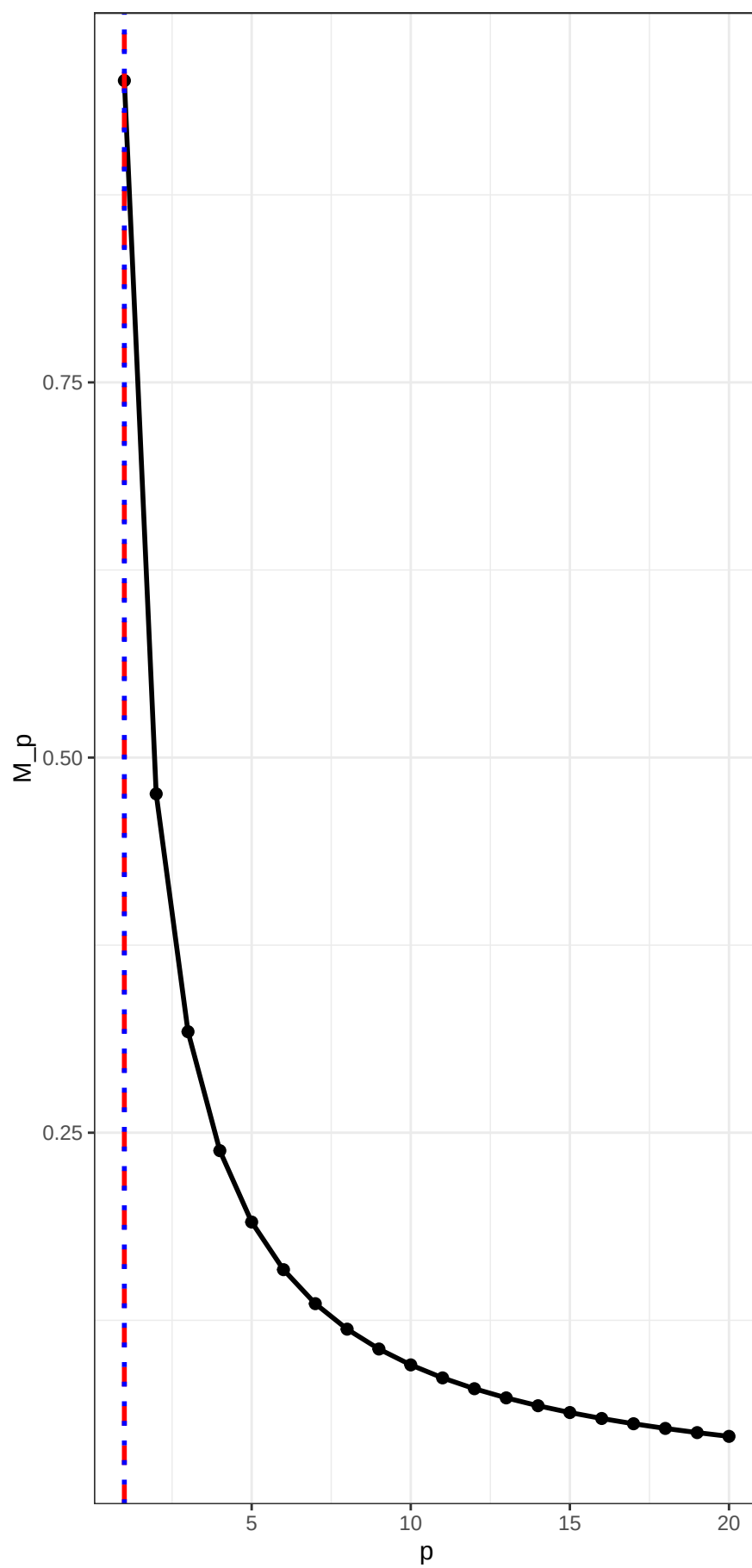
Third Derivative

$\text{argmin}(\cdot)=5$



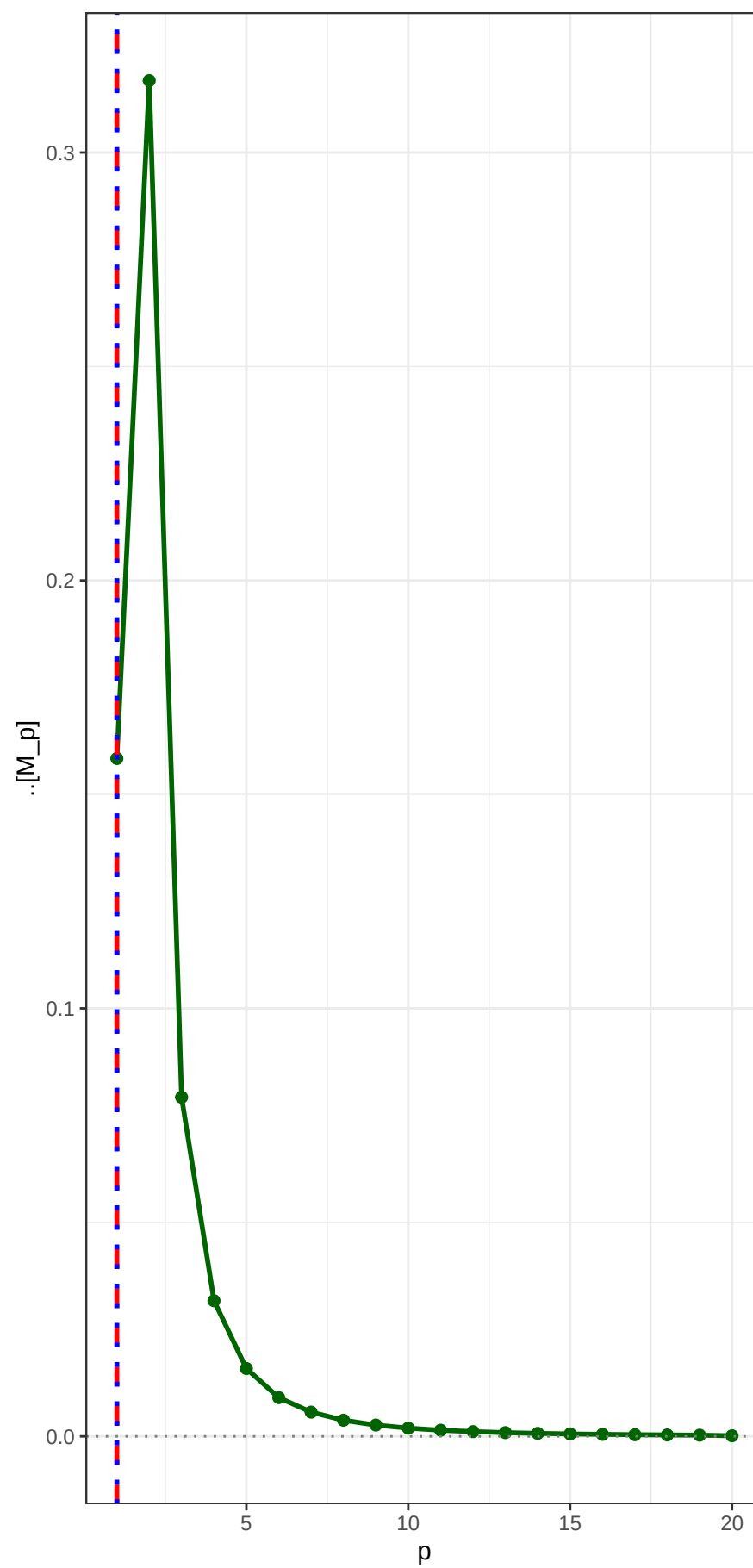
A2_Single_Predictor: $M_p = R^2/p$

$p_{\text{true}}=1$ (red), $p_{\text{classical}}=1$ (blue)



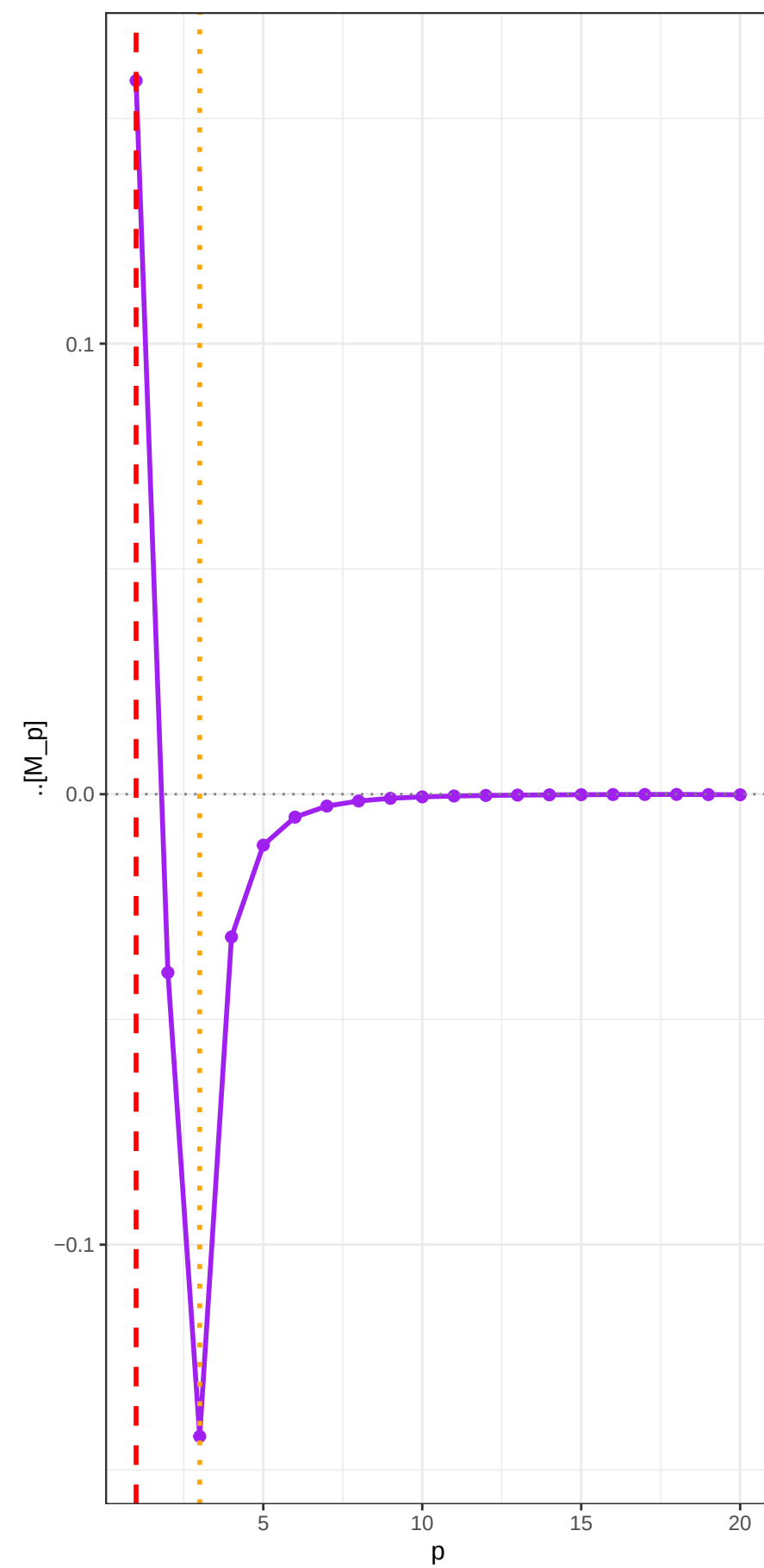
Second Derivative (Curvature)

$\Delta^2 > 0$ everywhere (M_p purely convex), maximum at $p=1$



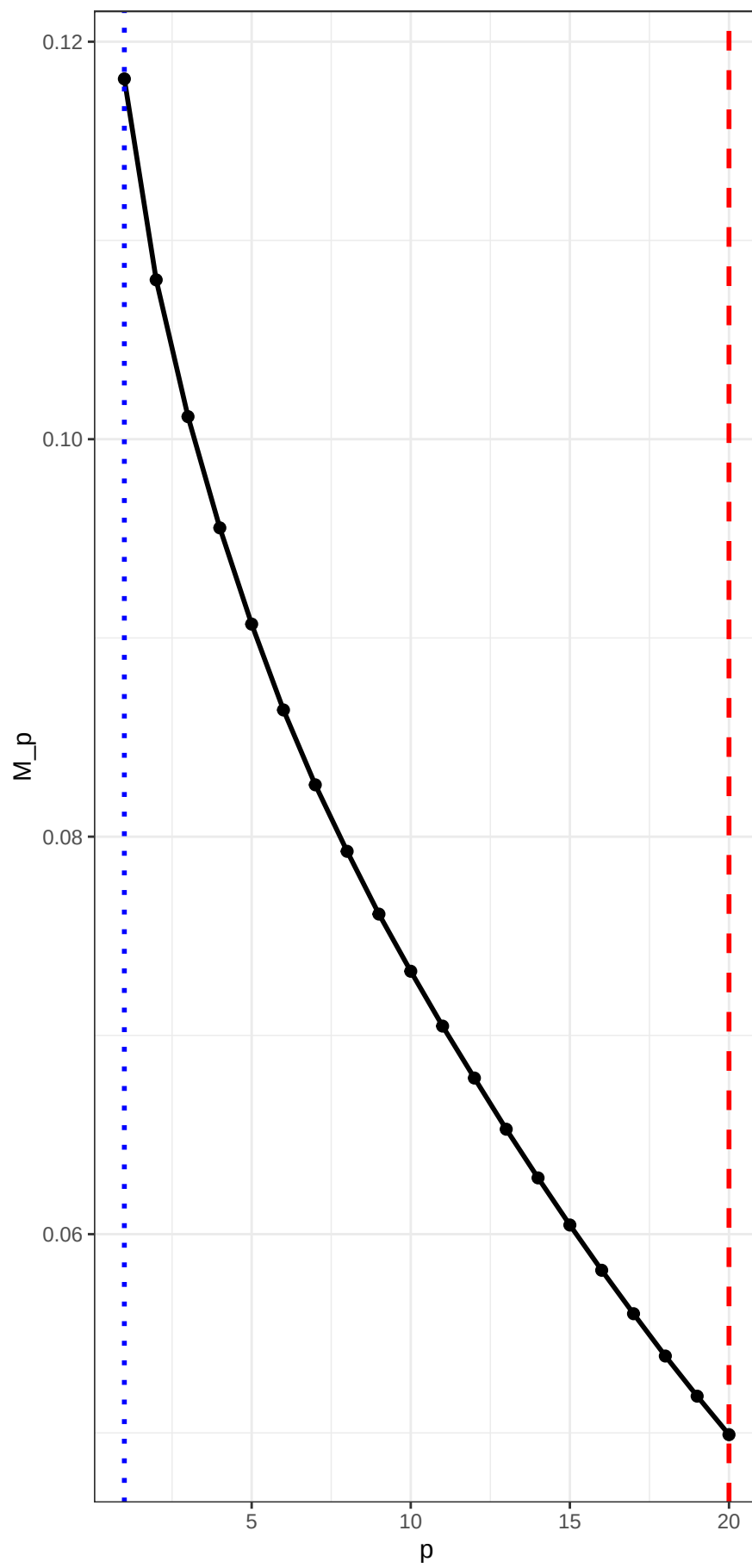
Third Derivative

$\text{argmin}(\cdot) = 3$



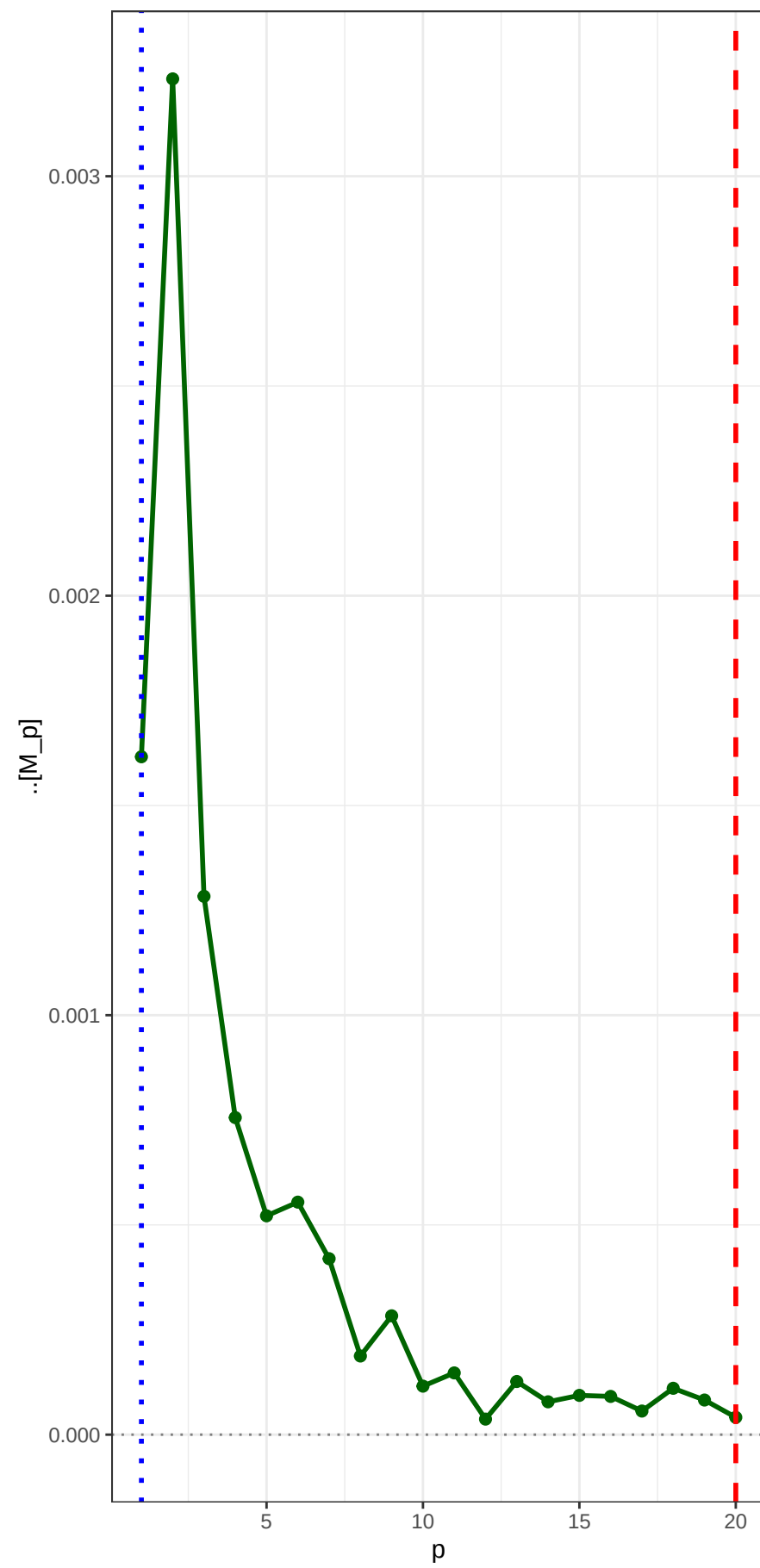
A3_Full_Support: $M_p = R^2/p$

$p_{\text{true}}=20$ (red), $p_{\text{classical}}=1$ (blue)



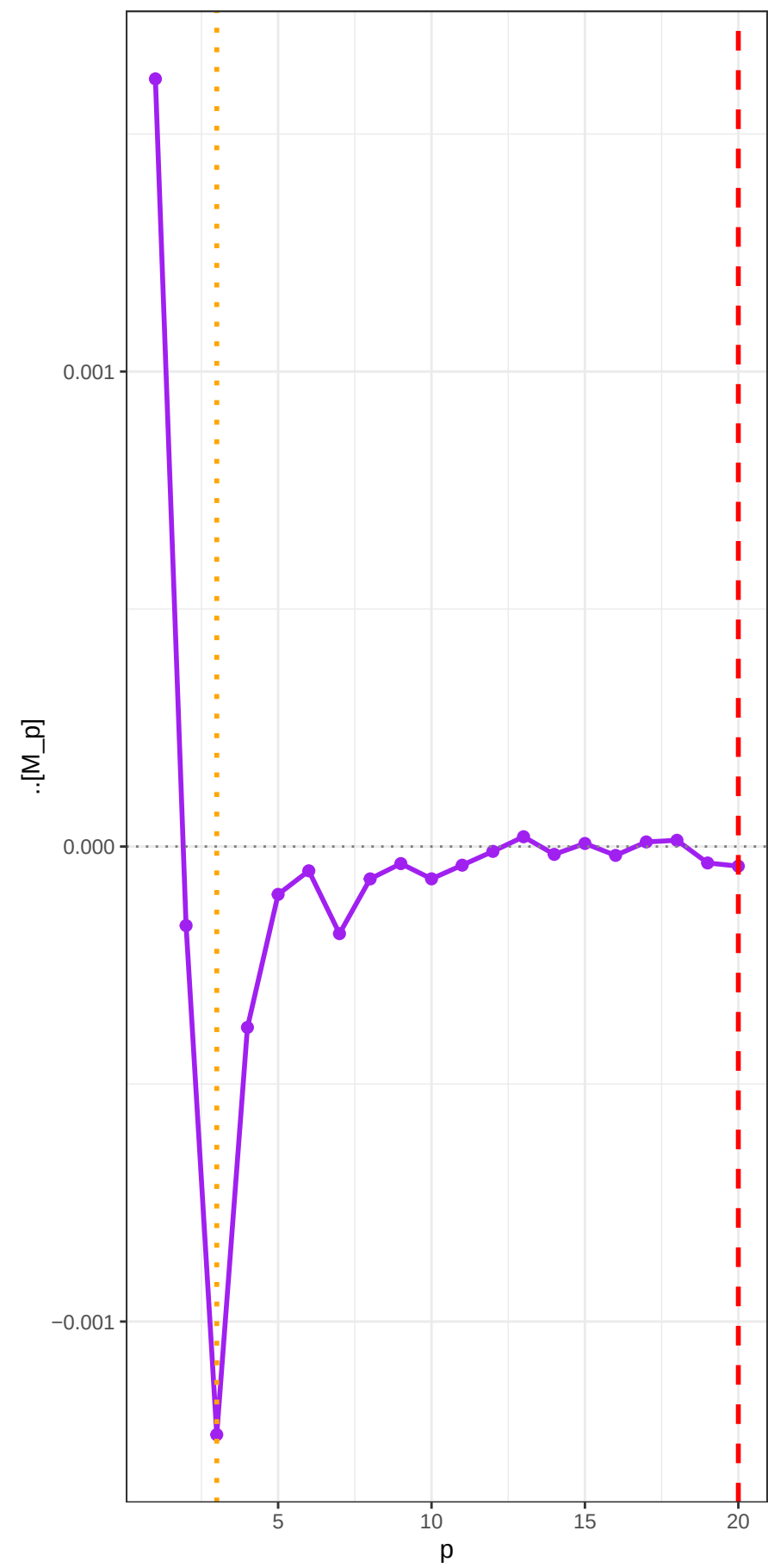
Second Derivative (Curvature)

$\Delta^2 > 0$ everywhere (M_p purely convex), maximum at $p=$

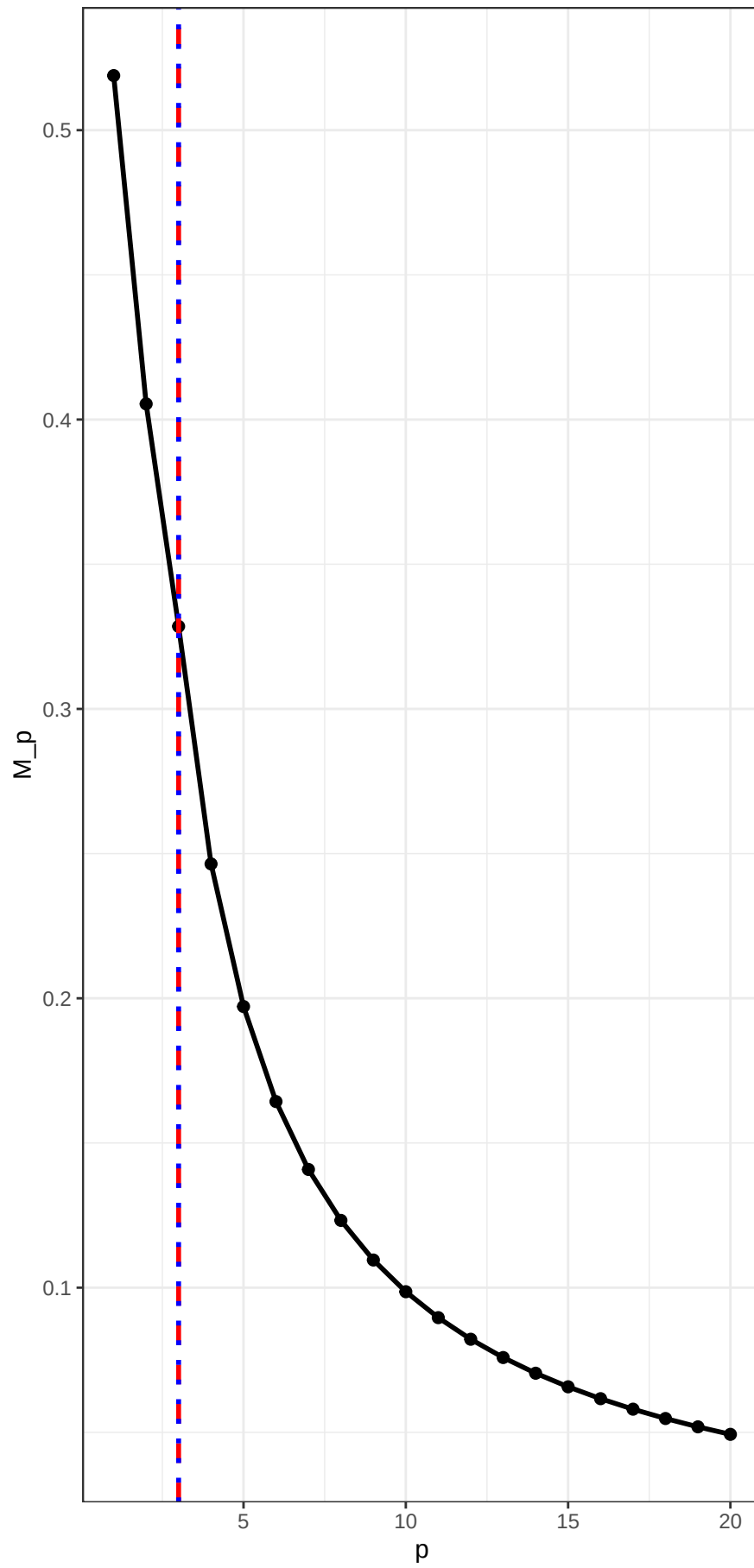


Third Derivative

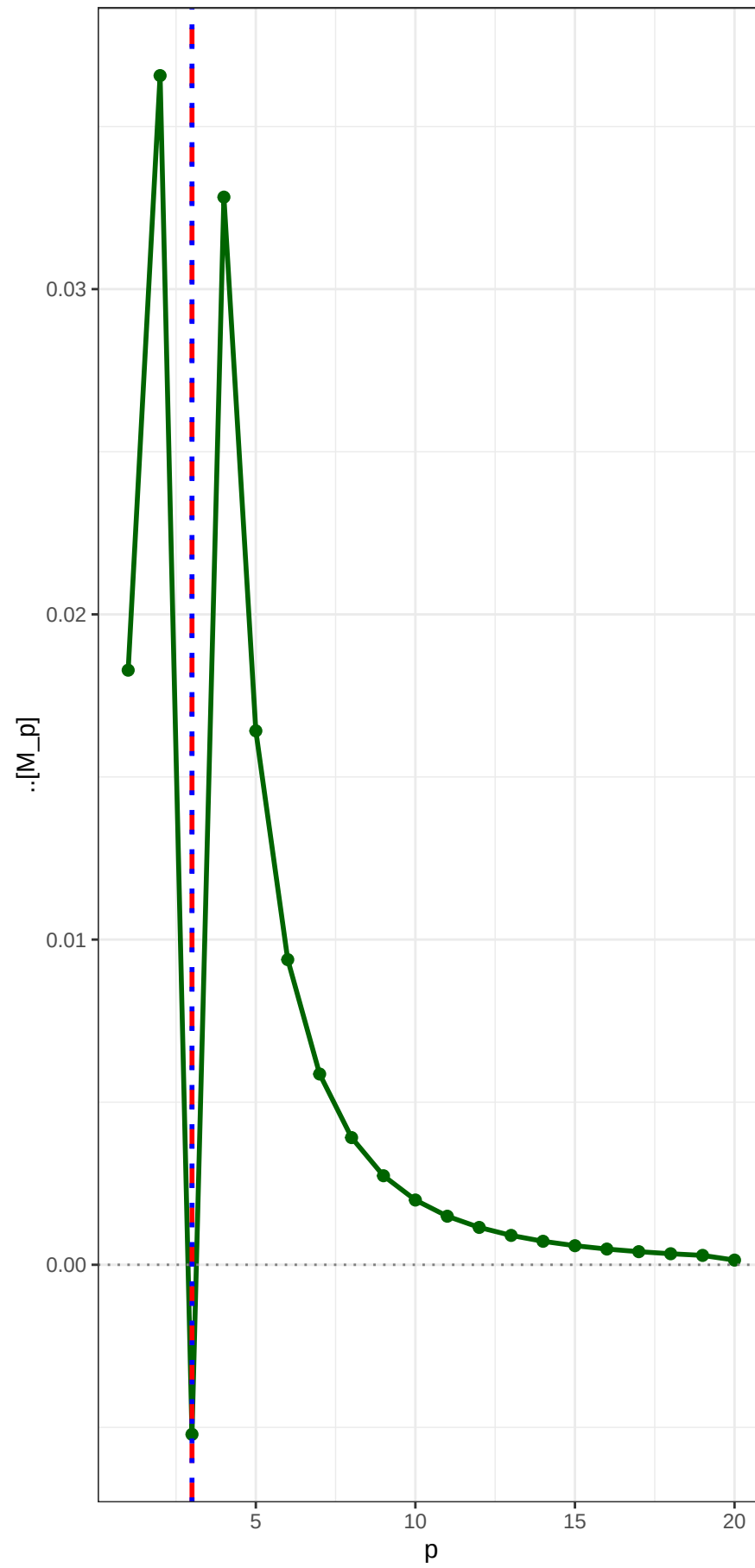
$\text{argmin}(\Delta^3)=3$



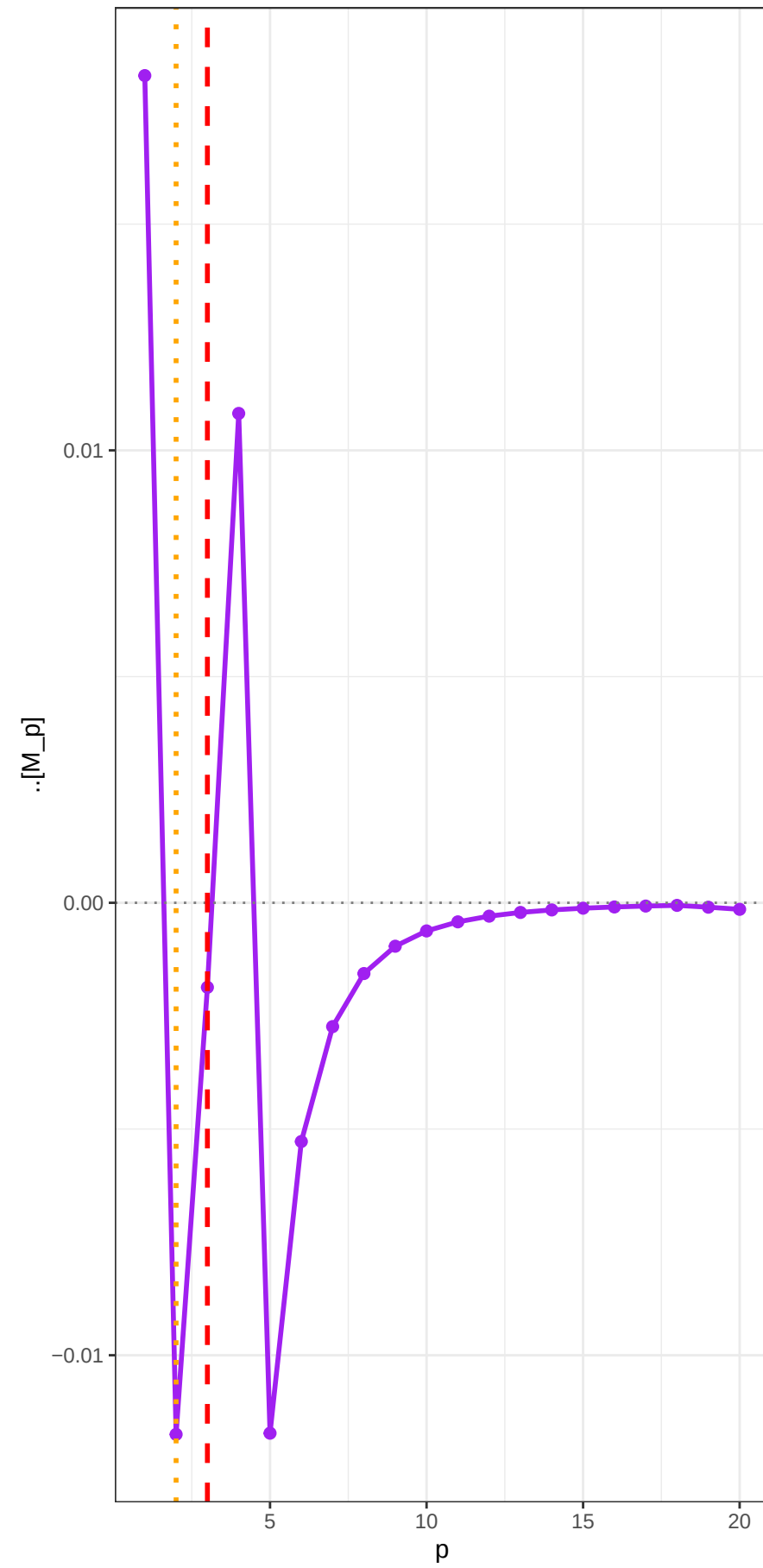
B1_AR1_Weak: $M_p = R^2/p$
 $p_{\text{true}}=3$ (red), $p_{\text{classical}}=3$ (blue)



Second Derivative (Curvature)
Sign change: $\text{delta2}[2]=0.03656647 \rightarrow \text{delta2}[3]=-0.005214$

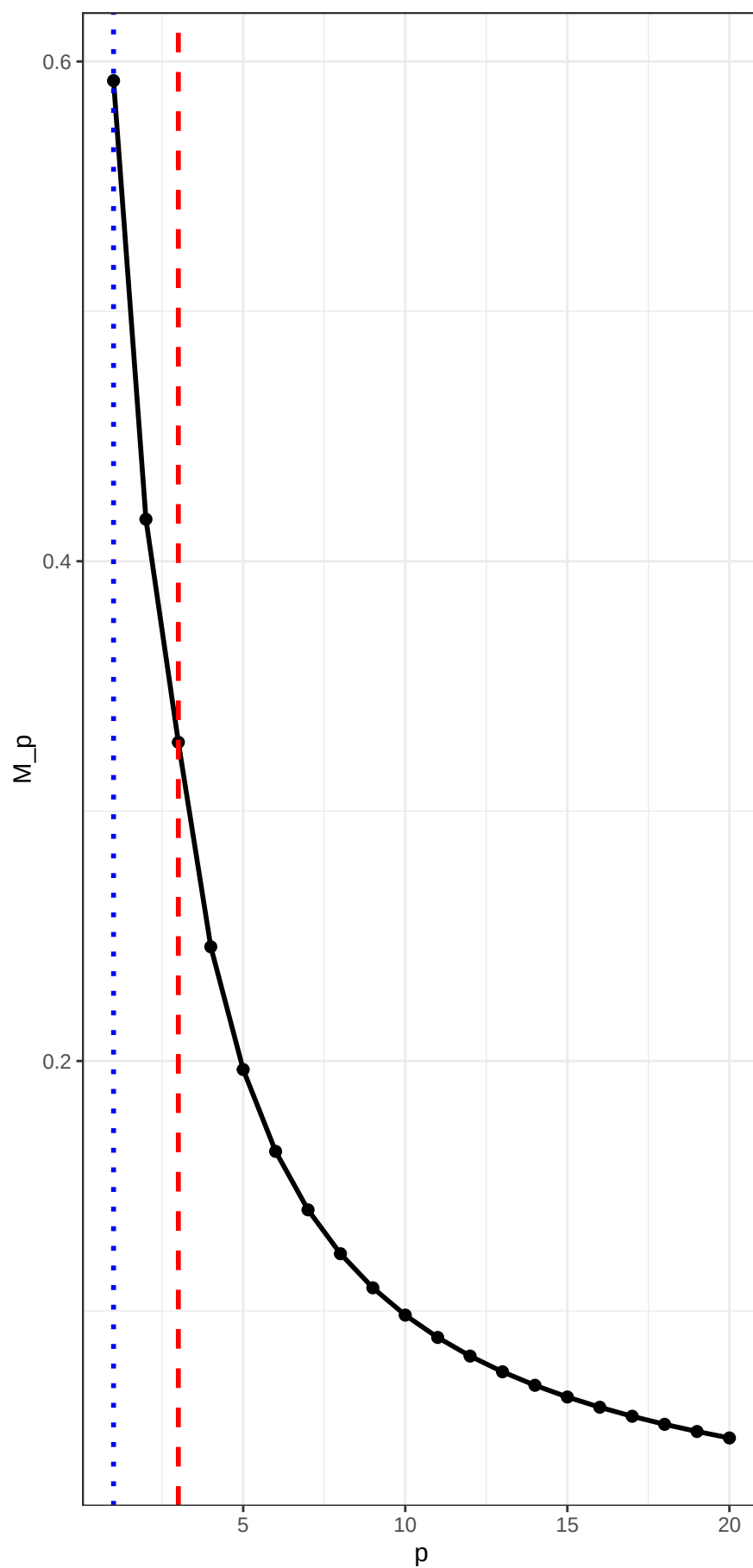


Third Derivative
 $\text{argmin}(\cdot)=2$



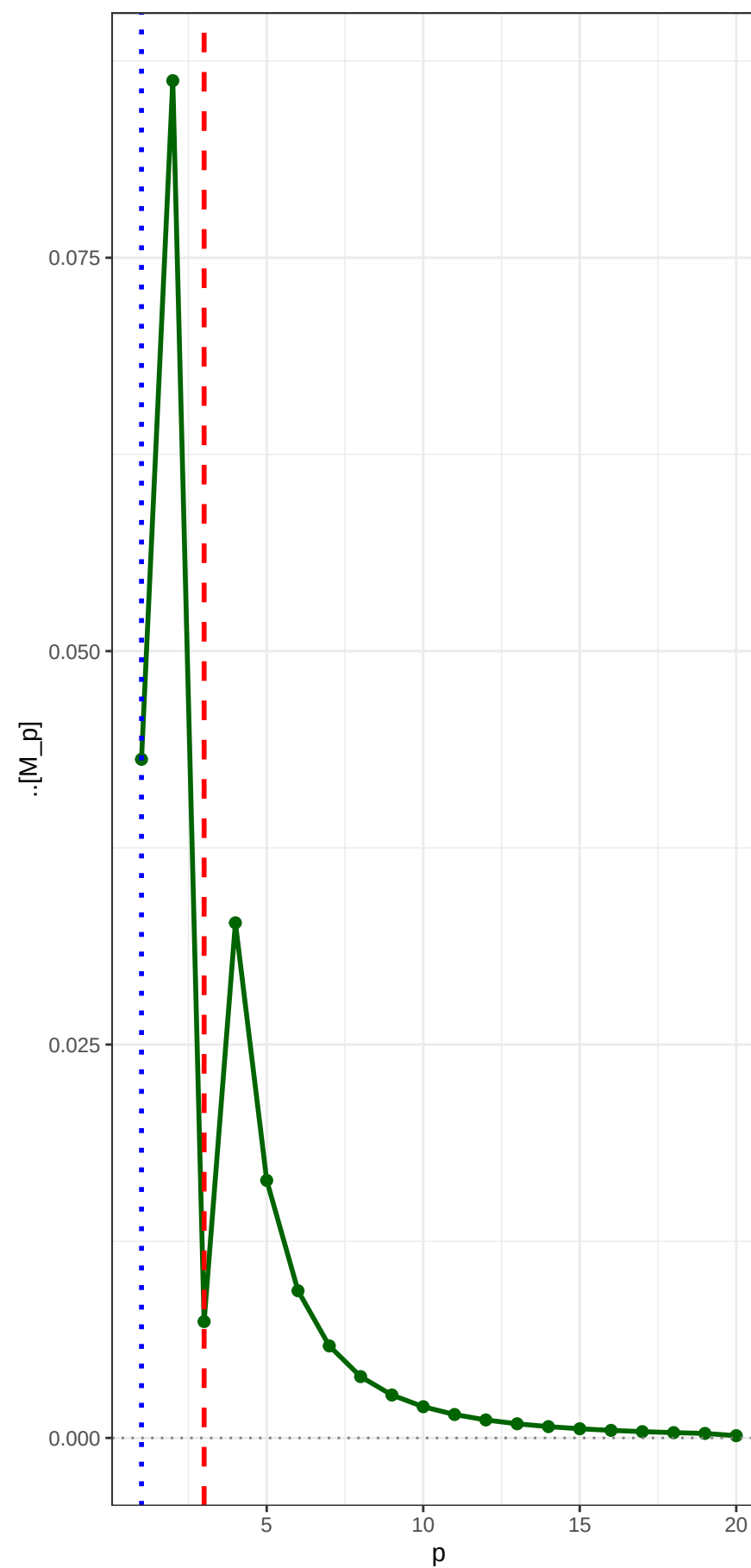
B1_AR1_Strong: $M_p = R^2/p$

$p_{\text{true}}=3$ (red), $p_{\text{classical}}=1$ (blue)



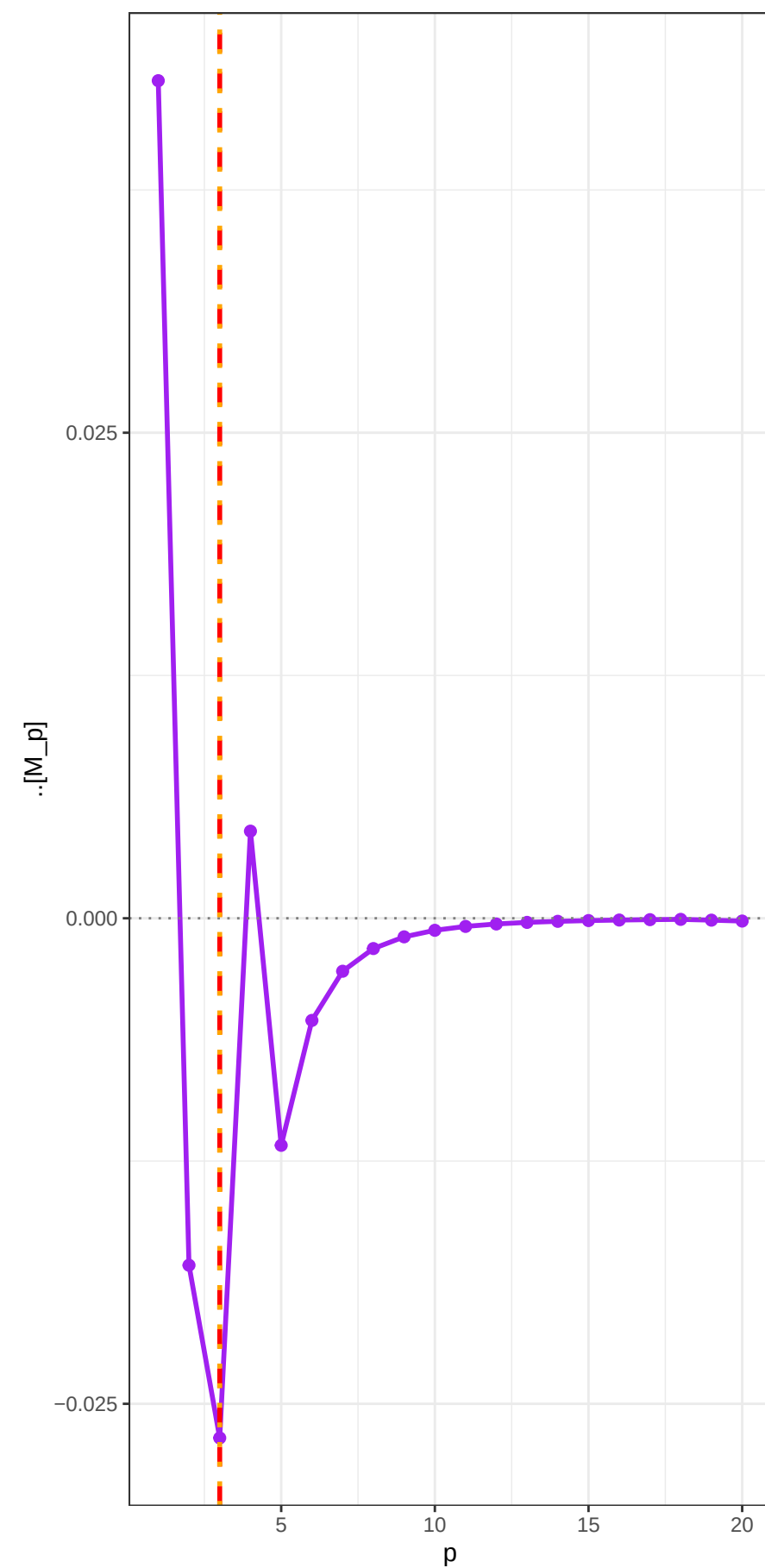
Second Derivative (Curvature)

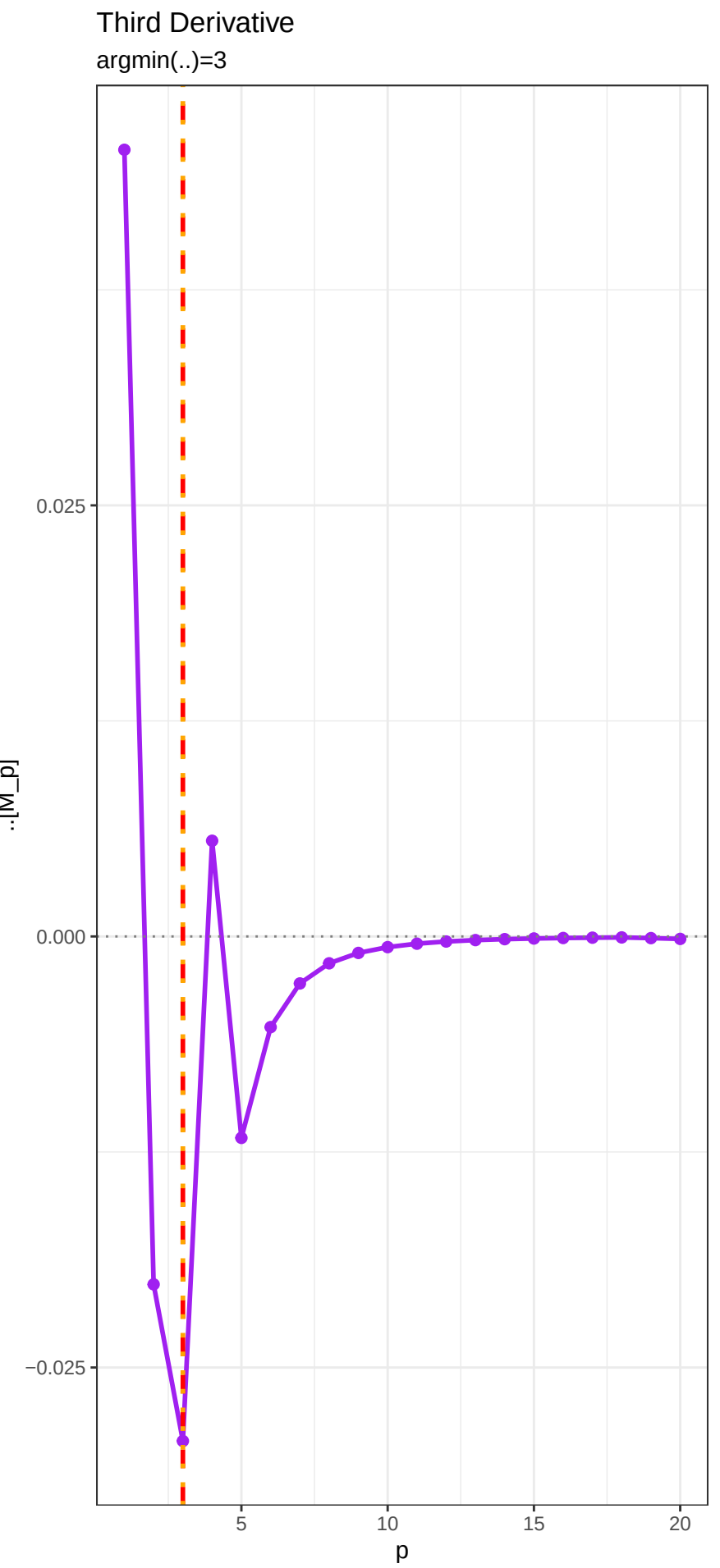
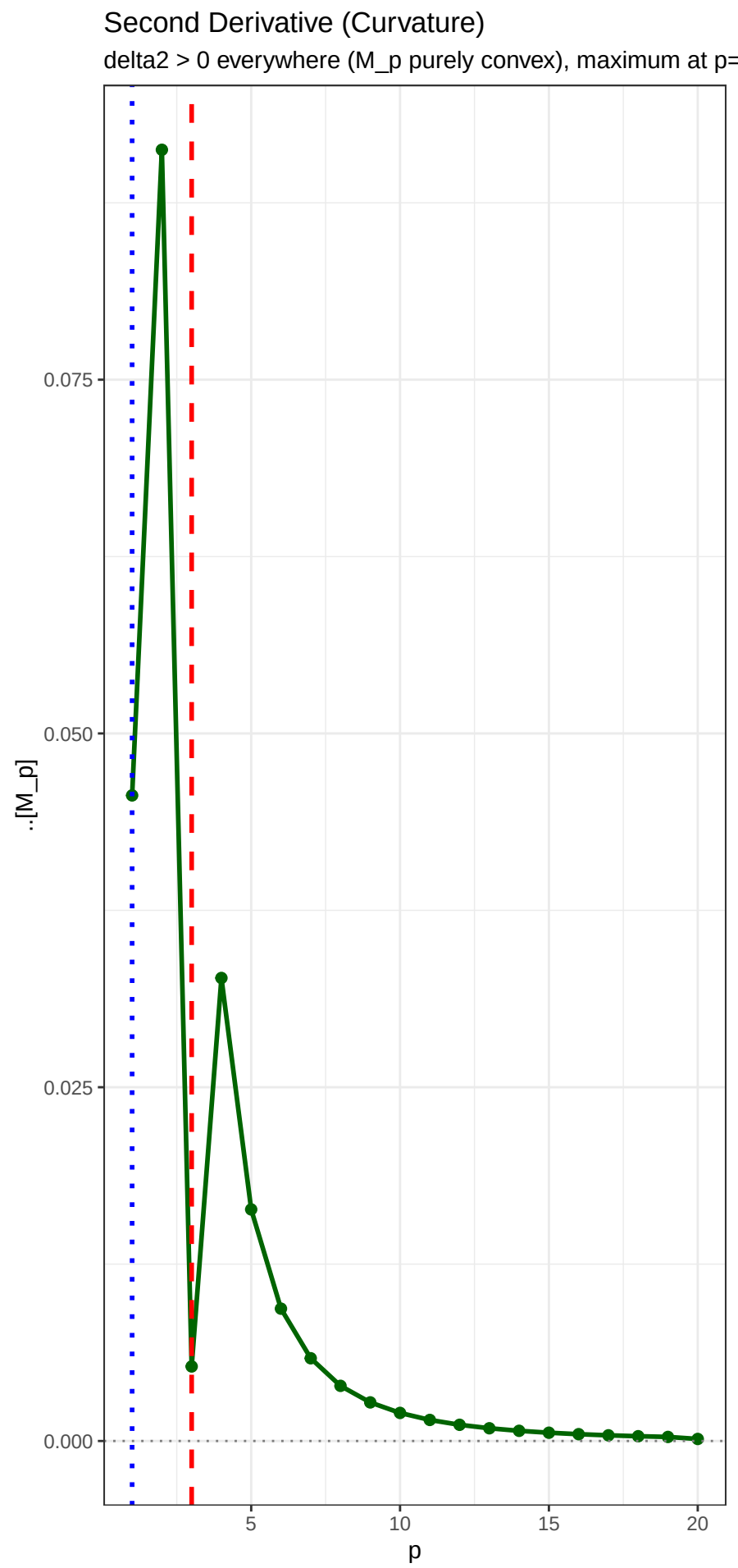
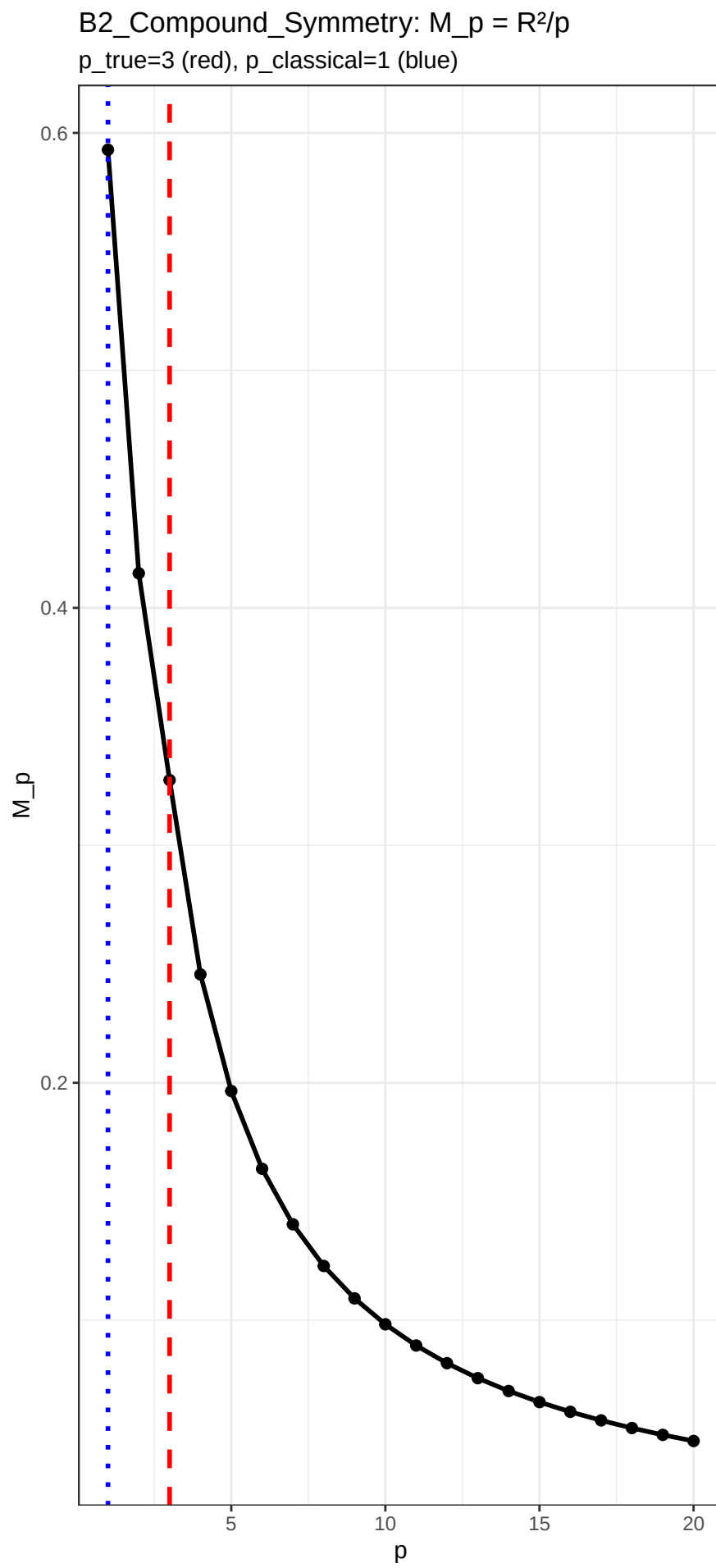
$\Delta^2 > 0$ everywhere (M_p purely convex), maximum at $p=$

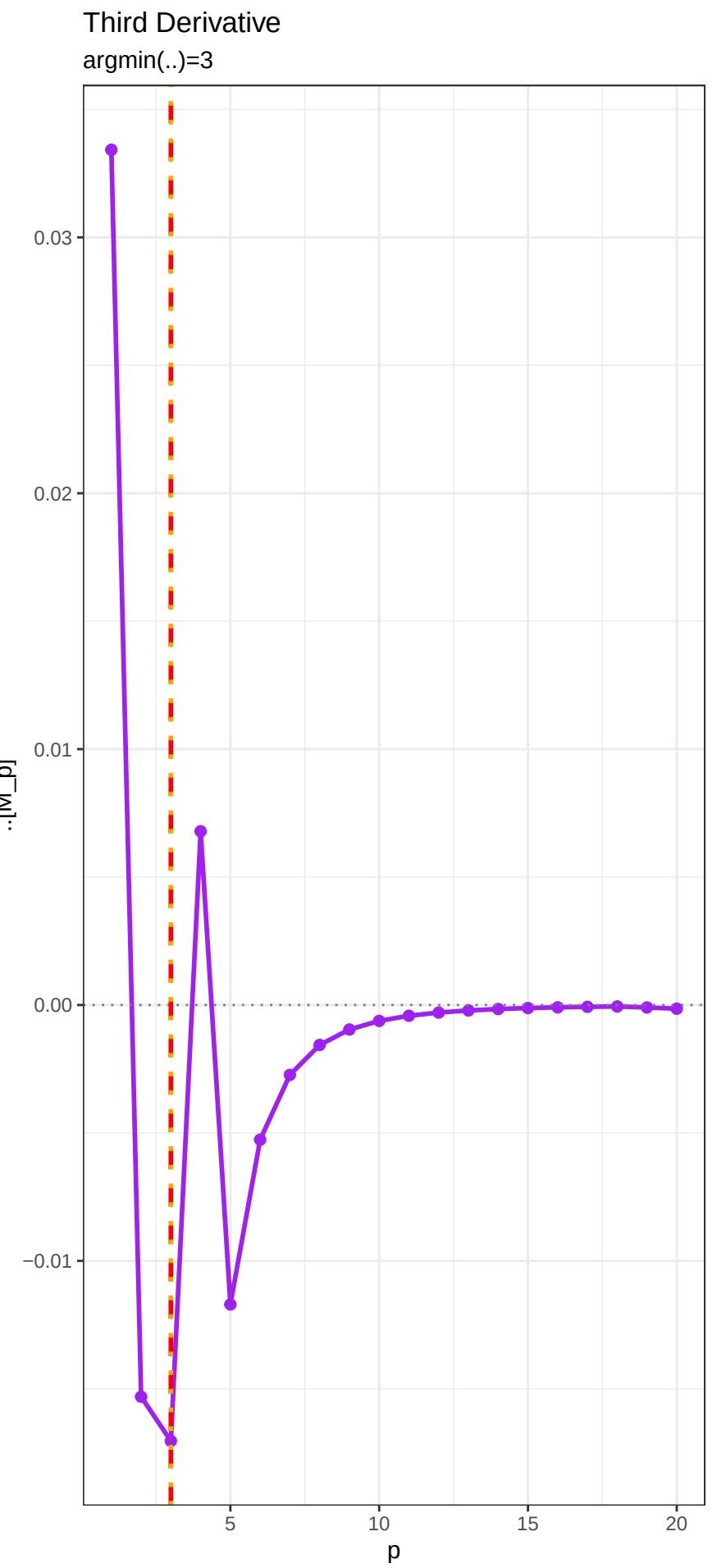
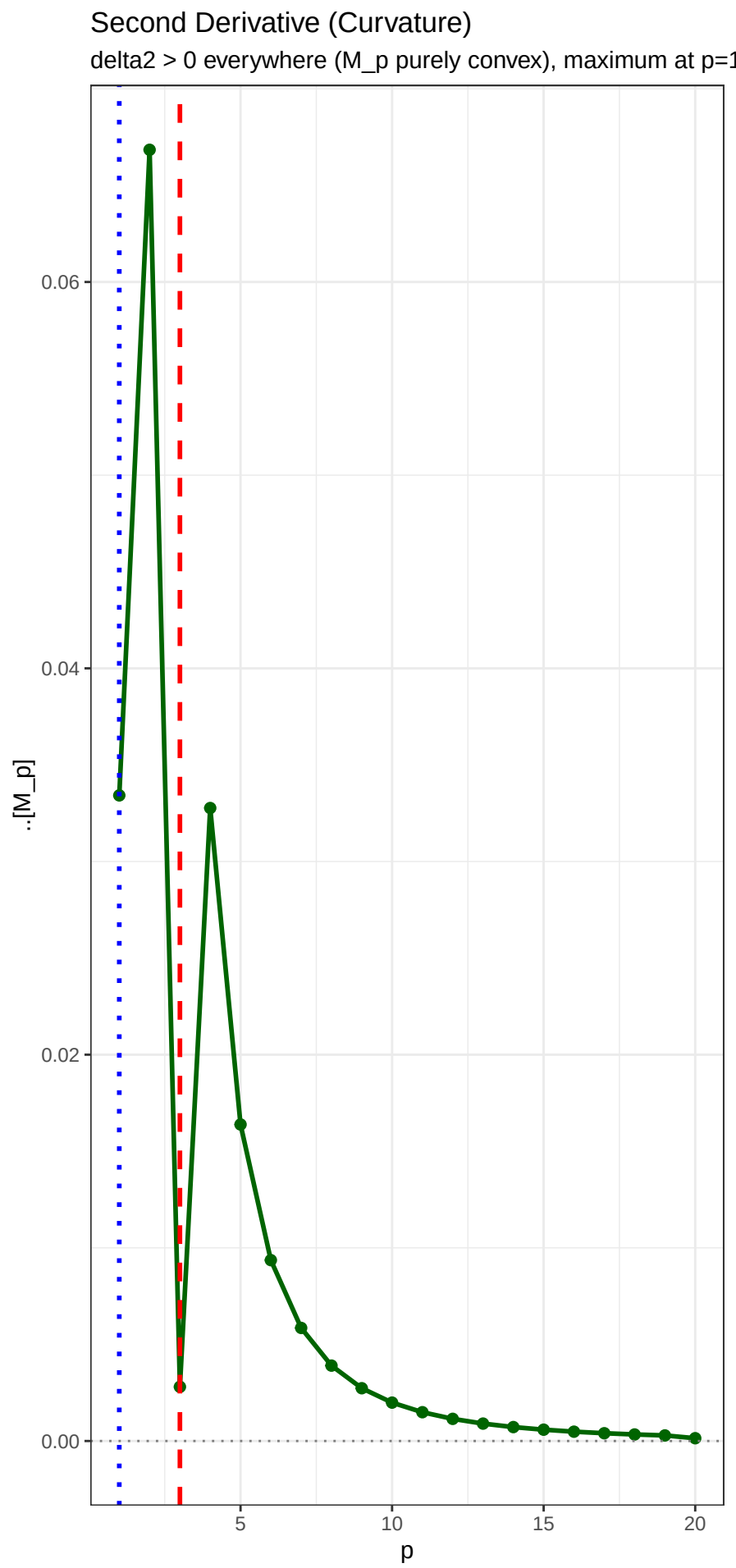
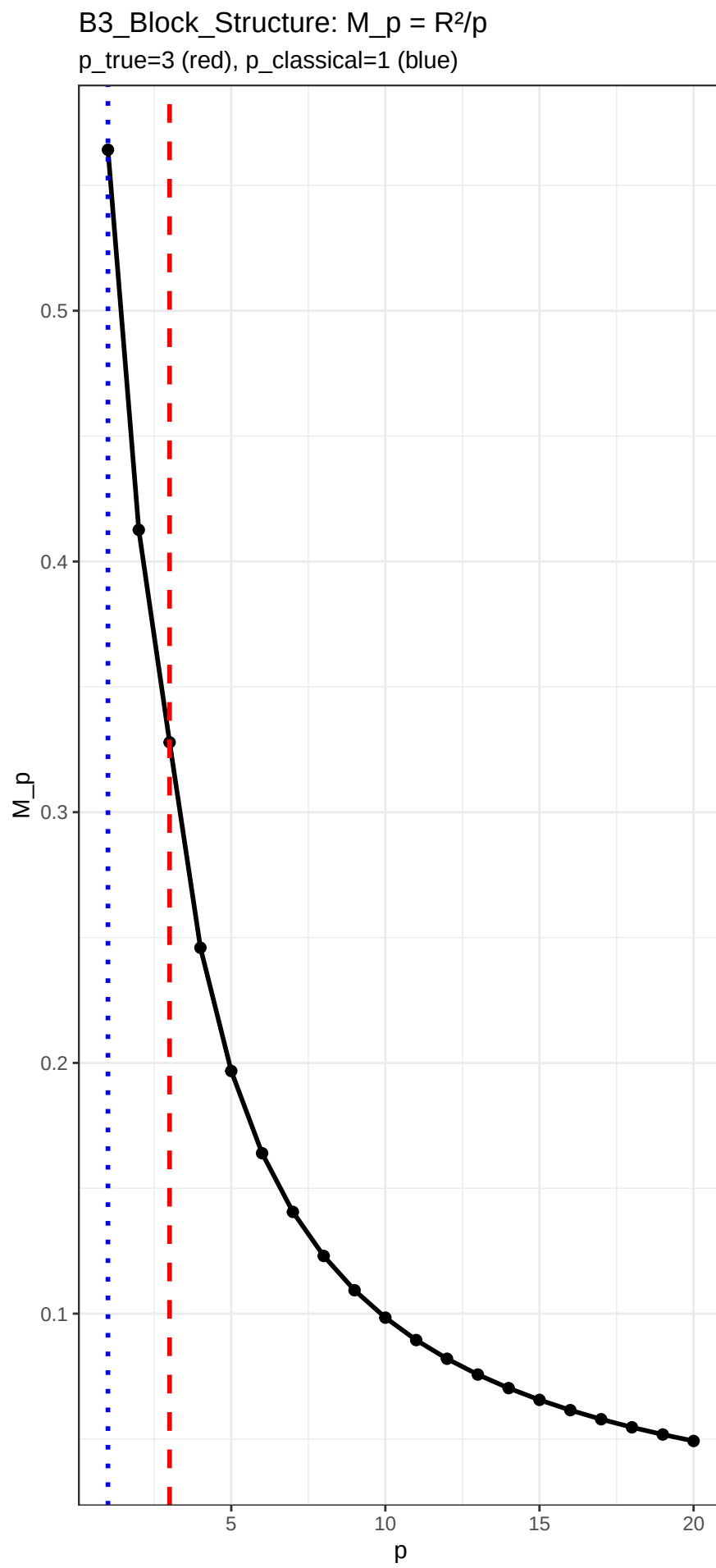


Third Derivative

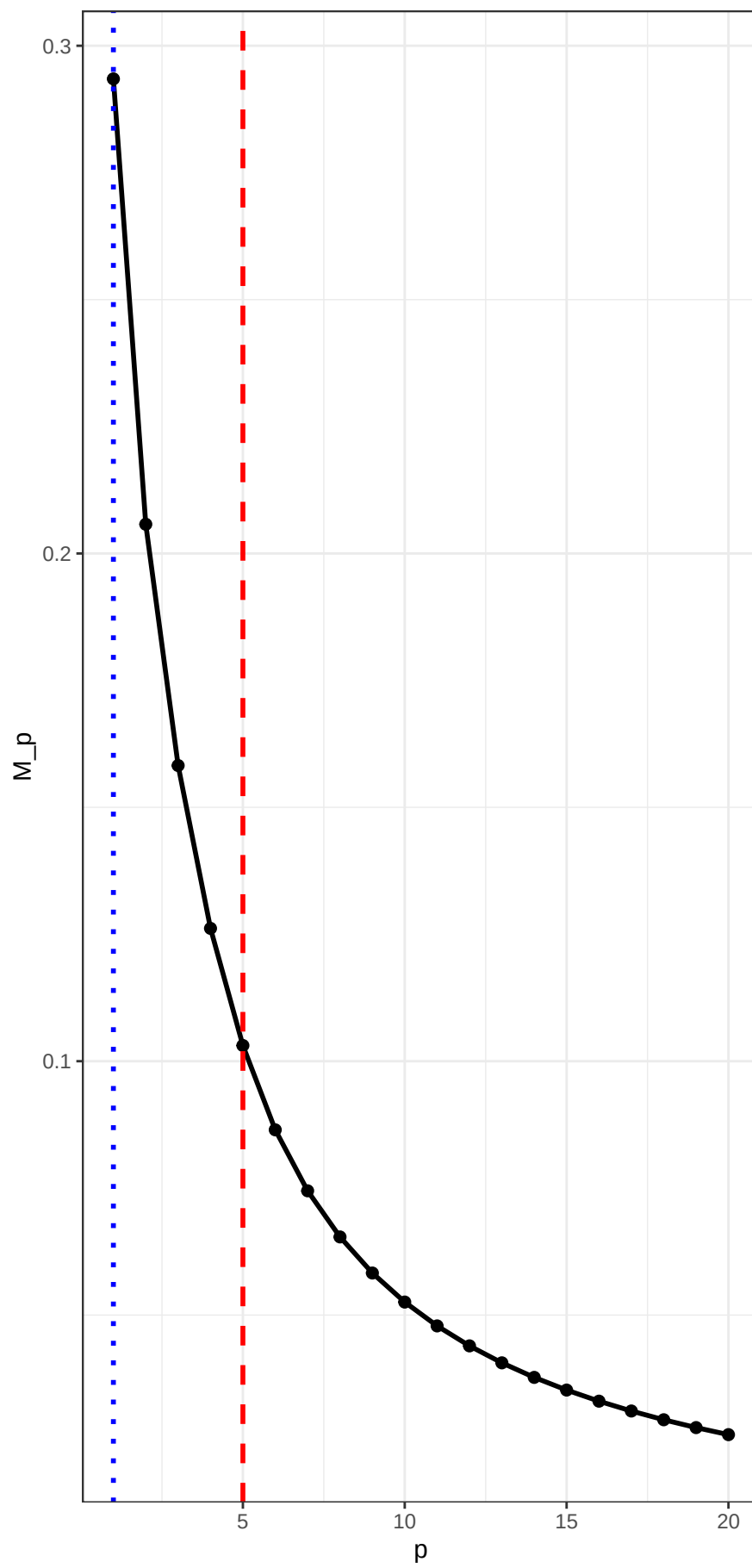
$\text{argmin}(\cdot)=3$



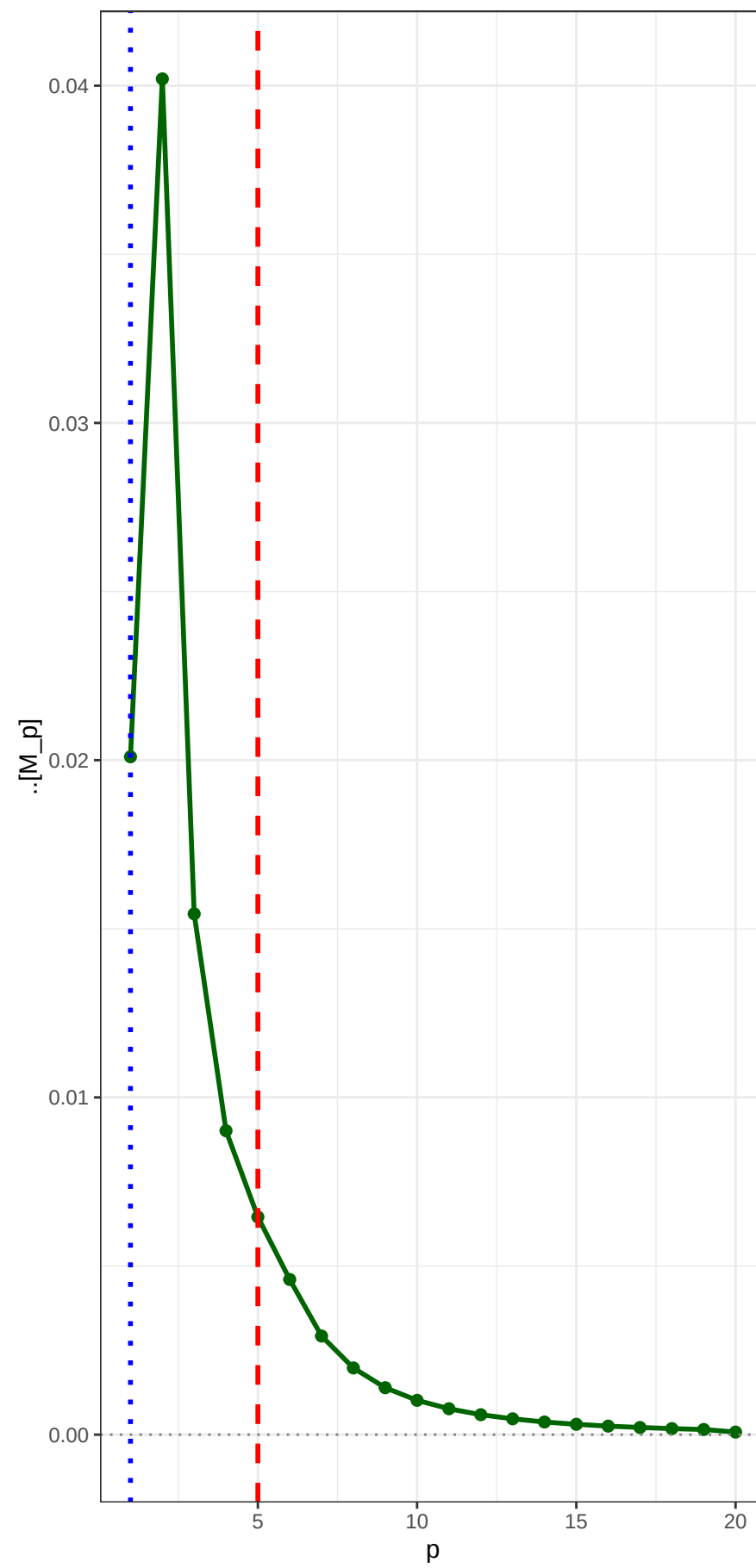




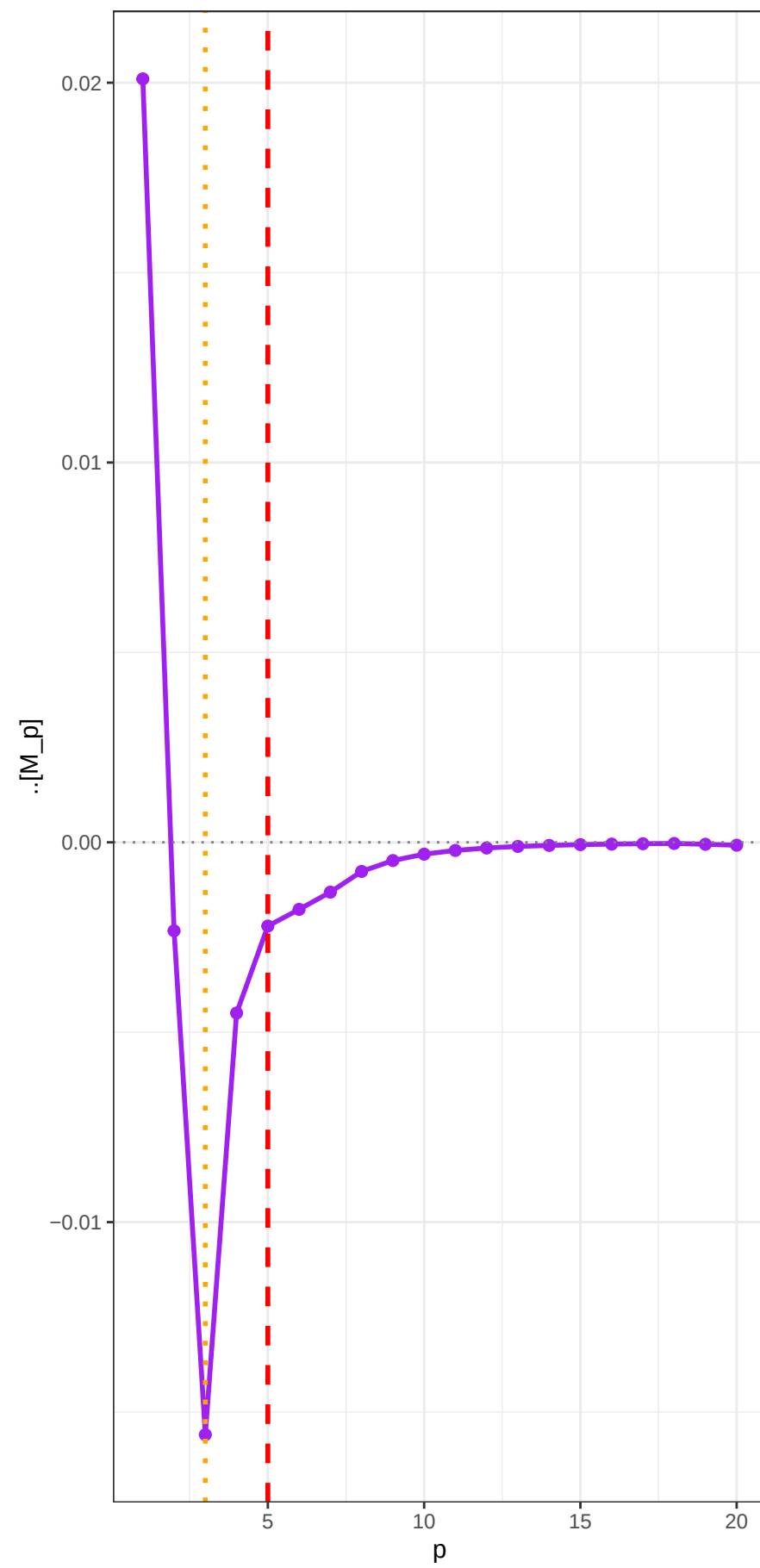
C1_Weak_Signals: $M_p = R^2/p$
 $p_{\text{true}}=5$ (red), $p_{\text{classical}}=1$ (blue)

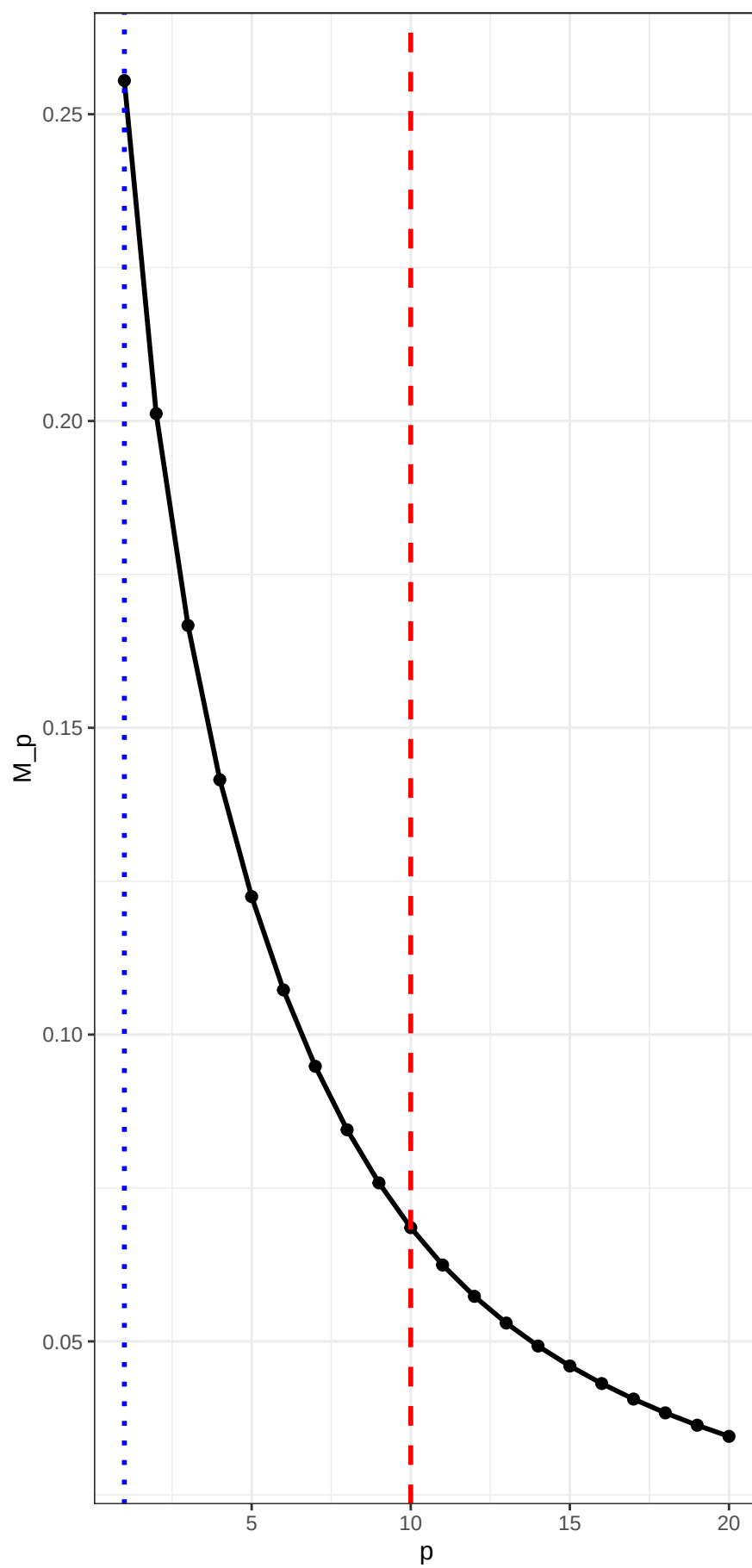


Second Derivative (Curvature)
 $\Delta^2 > 0$ everywhere (M_p purely convex), maximum at $p=1$

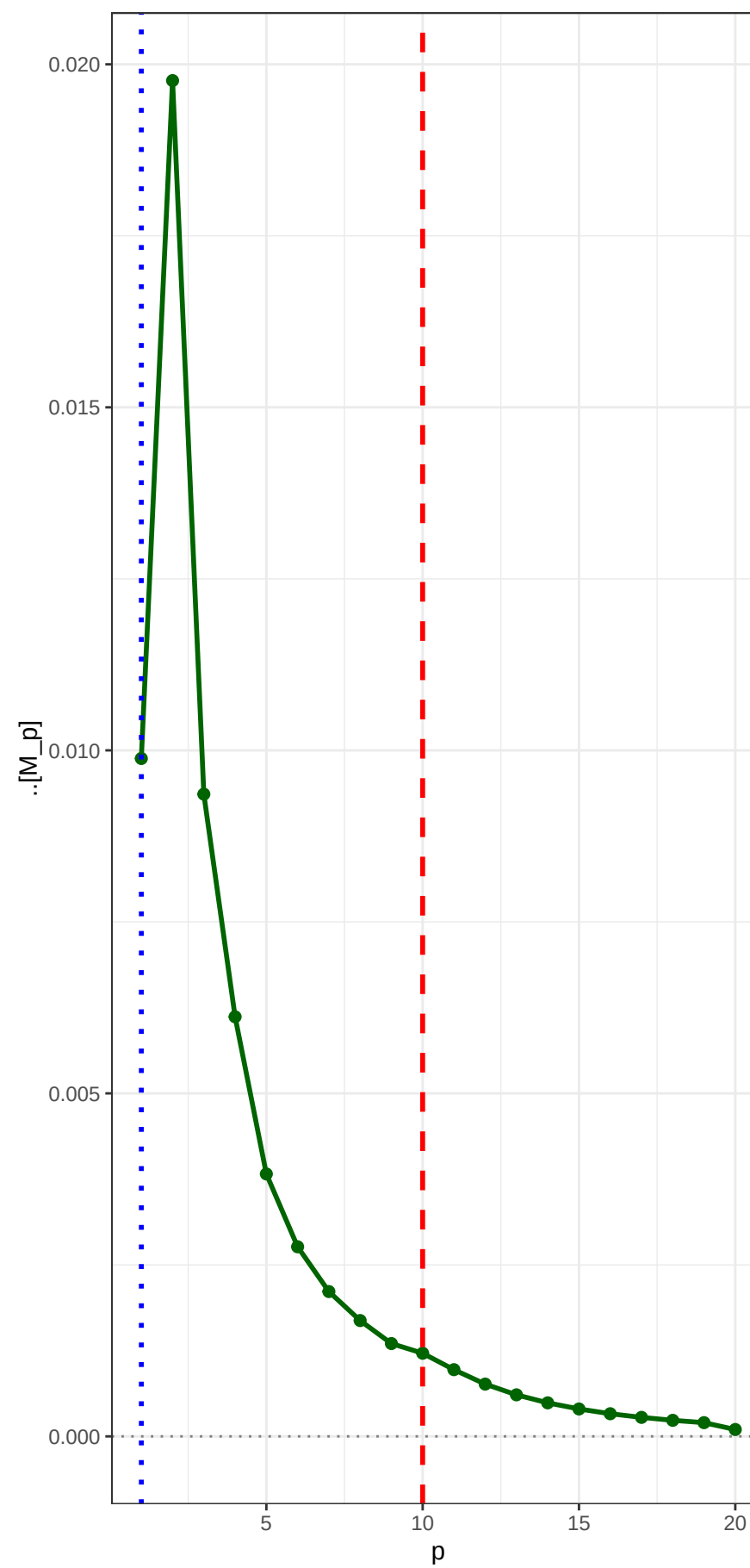


Third Derivative
 $\text{argmin}(\ddot{M}_p)=3$

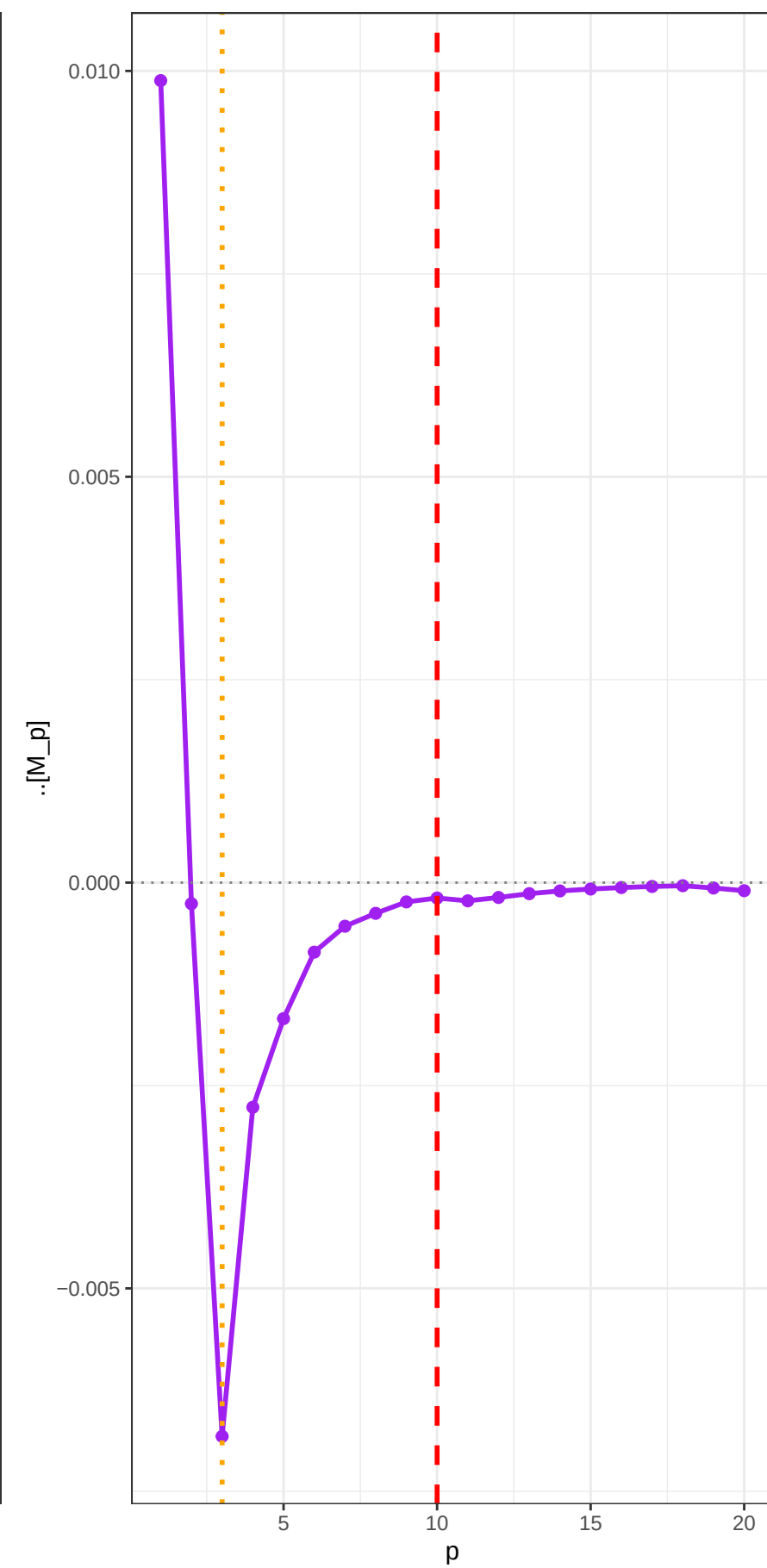


C2_Many_Weak_Signals: $M_p = R^2/p$ $p_{\text{true}}=10$ (red), $p_{\text{classical}}=1$ (blue)

Second Derivative (Curvature)

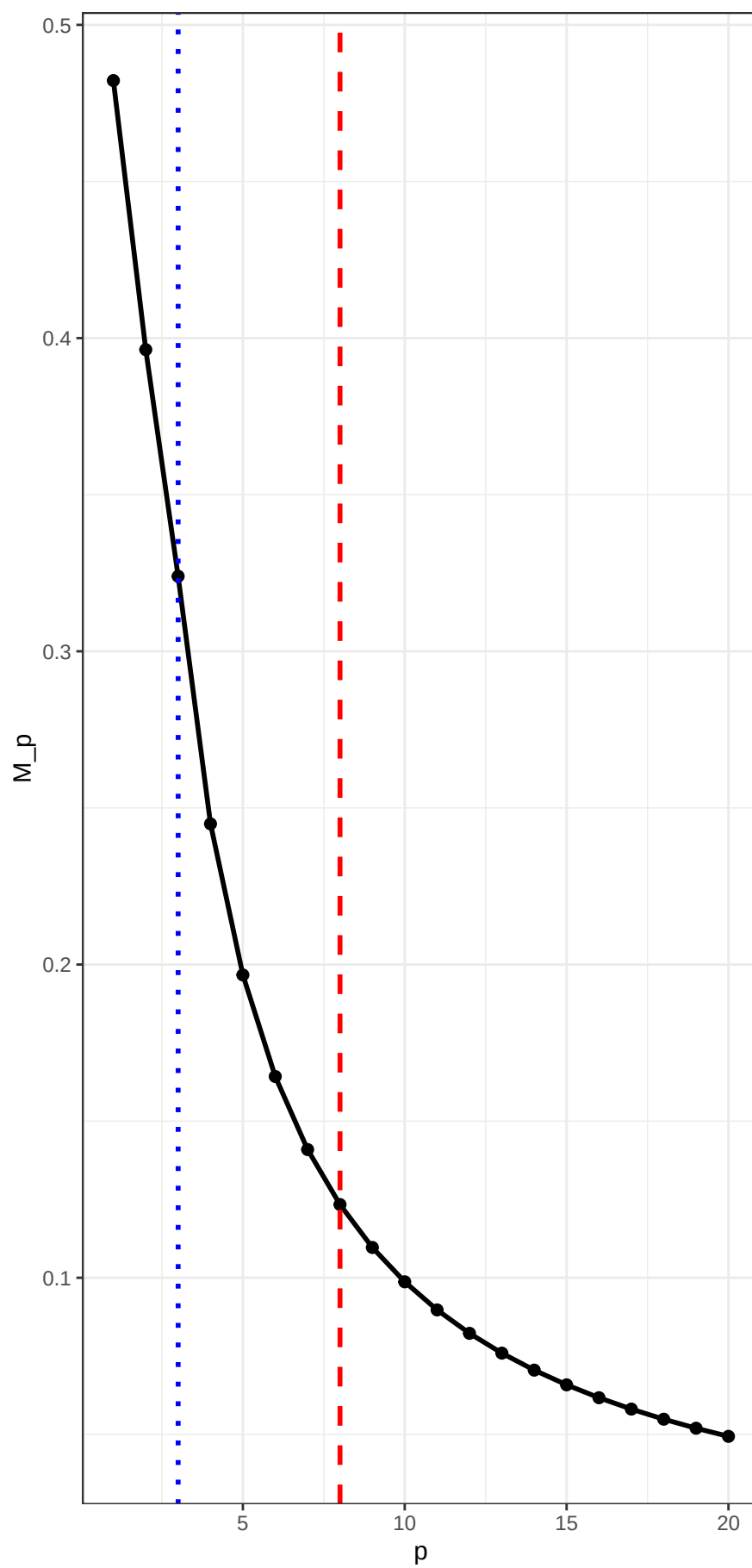
 $\Delta^2 > 0$ everywhere (M_p purely convex), maximum at $p=$ 

Third Derivative

 $\text{argmin}(\ddot{M}_p)=3$ 

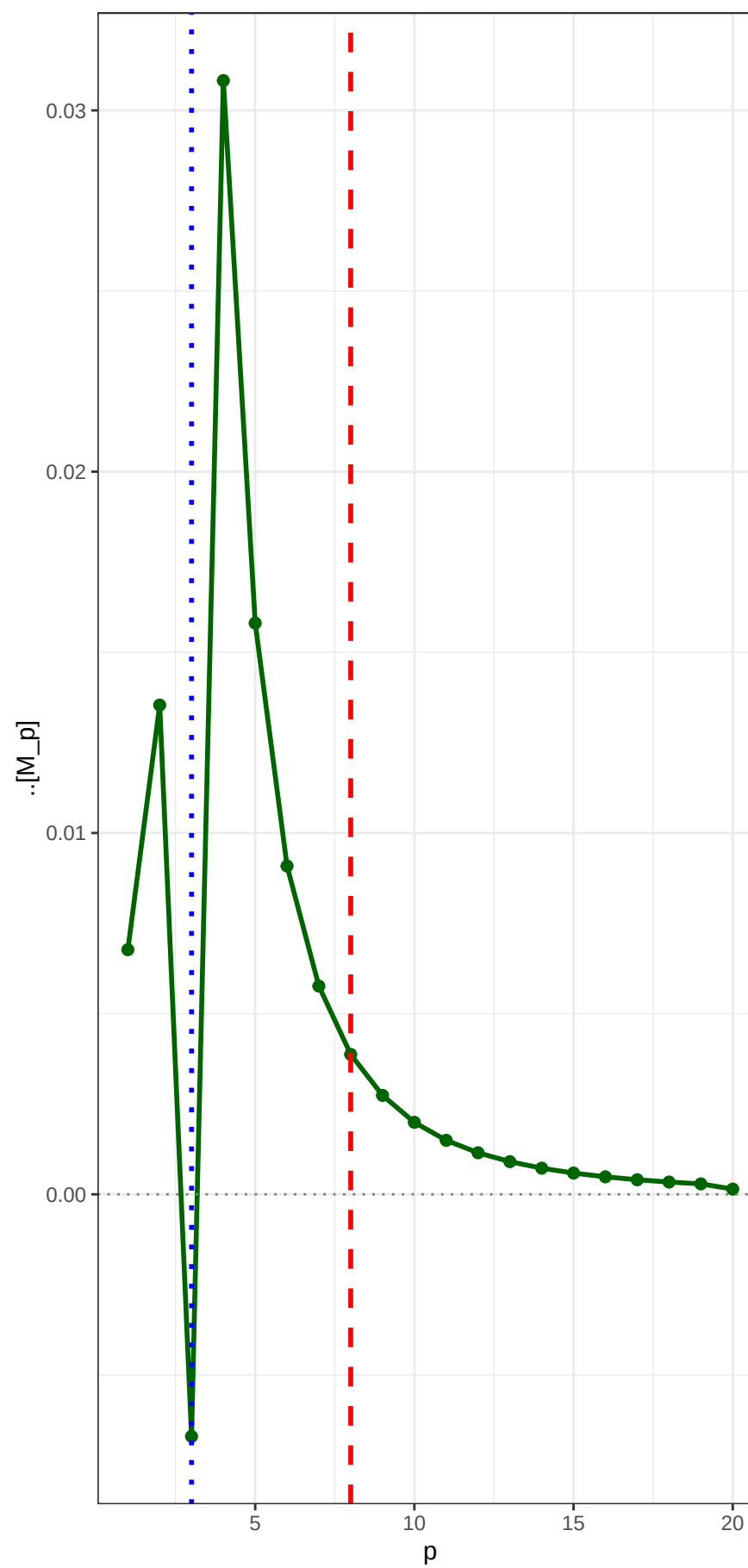
C3_Mixed_Signals: $M_p = R^2/p$

$p_{\text{true}}=8$ (red), $p_{\text{classical}}=3$ (blue)



Second Derivative (Curvature)

Sign change: $\Delta^2[2]=0.01354056 \rightarrow \Delta^2[3]=-0.006699$



Third Derivative

$\text{argmin}(\cdot)=5$

